

Exercises with Detailed Solutions

Kept with the original numeration of the book

Chapter 1: Exercises 1.1–1.6 with Solutions

1.1

The following measurements were recorded for the drying time, in hours, of a certain brand of latex paint:

3.4, 2.5, 4.8, 2.9, 3.6, 2.8, 3.3, 5.6, 3.7, 2.8, 4.4, 4.0, 5.2, 3.0, 4.8.

Assume that the measurements are a simple random sample.

- What is the sample size for the above sample?
- Calculate the sample mean for these data.
- Calculate the sample median.
- Plot the data by way of a dot plot.
- Compute the 20% trimmed mean for the above data set.
- Is the sample mean for these data more or less descriptive as a center of location than the trimmed mean?

Solution

- The sample contains 15 observations, so

$$n = 15.$$

- The sample mean is

$$\bar{x} = \frac{3.4 + 2.5 + 4.8 + 2.9 + 3.6 + 2.8 + 3.3 + 5.6 + 3.7 + 2.8 + 4.4 + 4.0 + 5.2 + 3.0 + 4.8}{15} = \frac{56.8}{15} = 3.786\bar{6}$$

Hence

$$\bar{x} \approx 3.787.$$

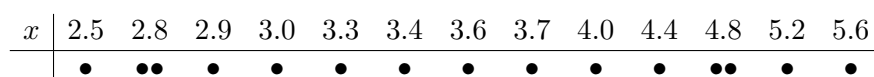
- Order the data:

2.5, 2.8, 2.8, 2.9, 3.0, 3.3, 3.4, 3.6, 3.7, 4.0, 4.4, 4.8, 4.8, 5.2, 5.6.

Since $n = 15$, the median is the $\frac{15+1}{2} = 8$ th value:

$$\tilde{x} = 3.6.$$

- A simple text dot plot is



- (e) For a 20% trimmed mean with $n = 15$,

$$0.20(15) = 3,$$

so remove the lowest 3 and highest 3 observations.

Ordered data:

2.5, 2.8, 2.8, 2.9, 3.0, 3.3, 3.4, 3.6, 3.7, 4.0, 4.4, 4.8, 4.8, 5.2, 5.6.

Remove

2.5, 2.8, 2.8 and 4.8, 5.2, 5.6.

Remaining values:

2.9, 3.0, 3.3, 3.4, 3.6, 3.7, 4.0, 4.4, 4.8.

Their sum is

$$33.1,$$

so the trimmed mean is

$$\bar{x}_t = \frac{33.1}{9} = 3.6778.$$

Thus

$$\boxed{\bar{x}_{0.20} \approx 3.678.}$$

- (f) The sample mean is pulled upward somewhat by the larger values 5.2 and 5.6. The trimmed mean is less sensitive to these extremes, so it gives a slightly more resistant measure of center. Therefore,

$\boxed{\text{the 20\% trimmed mean is slightly more descriptive than the ordinary mean.}}$

1.2

According to the journal *Chemical Engineering*, an important property of a fiber is its water absorbency. A random sample of 20 pieces of cotton fiber was taken and the absorbency on each piece was measured. The data are:

18.71, 21.41, 20.72, 21.81, 19.29, 22.43, 20.17,

23.71, 19.44, 20.50, 18.92, 20.33, 23.00, 22.85,

19.25, 21.77, 22.11, 19.77, 18.04, 21.12.

- Calculate the sample mean and median for the above sample values.
- Compute the 10% trimmed mean.
- Do a dot plot of the absorbency data.
- Using only the values of the mean, median, and trimmed mean, do you have evidence of outliers in the data?

Solution

- (a) The sample mean is

$$\bar{x} = \frac{415.35}{20} = 20.7675.$$

Hence

$$\boxed{\bar{x} = 20.7675.}$$

Order the data:

18.04, 18.71, 18.92, 19.25, 19.29, 19.44, 19.77, 20.17, 20.33, 20.50,
 20.72, 21.12, 21.41, 21.77, 21.81, 22.11, 22.43, 22.85, 23.00, 23.71.

Since $n = 20$, the median is the average of the 10th and 11th values:

$$\tilde{x} = \frac{20.50 + 20.72}{2} = 20.61.$$

So

$$\boxed{\tilde{x} = 20.61.}$$

- (b) A 10% trimmed mean removes the lowest 2 and highest 2 observations.

Remove

18.04, 18.71 and 23.00, 23.71.

The remaining 16 values sum to

$$331.89,$$

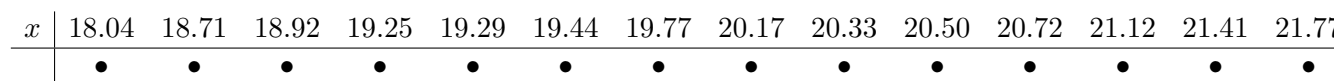
so

$$\bar{x}_{0.10} = \frac{331.89}{16} = 20.743125.$$

Thus

$$\boxed{\bar{x}_{0.10} \approx 20.743.}$$

- (c) A simple dot plot is



- (d) The mean, median, and trimmed mean are all very close:

$$20.7675, \quad 20.61, \quad 20.743.$$

Since these three measures are similar, there is no strong evidence of extreme outliers based only on them.

$$\boxed{\text{No strong evidence of outliers is suggested by these summary measures alone.}}$$

1.3

A certain polymer is used for evacuation systems for aircraft. It is important that the polymer be resistant to the aging process. Twenty specimens of the polymer were used in an experiment. Ten were assigned randomly to be exposed to an accelerated batch aging process that involved exposure to high temperatures for 10 days. Measurements of tensile strength of the specimens were made, and the following data were recorded:

No aging:

227, 222, 218, 217, 225, 218, 216, 229, 228, 221

Aging:

219, 214, 215, 211, 209, 218, 203, 204, 201, 205

- Do a dot plot of the data.
- From your plot, does it appear as if the aging process has had an effect on the tensile strength of this polymer? Explain.
- Calculate the sample mean tensile strength of the two samples.
- Calculate the sample median for both. Discuss the similarity or lack of similarity between the mean and median of each group.

Solution

- Ordered data:

No aging:

216, 217, 218, 218, 221, 222, 225, 227, 228, 229

Aging:

201, 203, 204, 205, 209, 211, 214, 215, 218, 219

A text dot plot clearly shows the aged sample shifted to the left of the non-aged sample.

- Yes. The aged values are generally much smaller than the non-aged values, so the plot suggests that aging lowers tensile strength.

The aging process appears to reduce tensile strength.

- Sample means:

For no aging,

$$\bar{x}_1 = \frac{2221}{10} = 222.1.$$

For aging,

$$\bar{x}_2 = \frac{2099}{10} = 209.9.$$

Therefore

$$\bar{x}_{\text{no aging}} = 222.1, \quad \bar{x}_{\text{aging}} = 209.9.$$

(d) Sample medians:

For no aging, the median is the average of the 5th and 6th ordered values:

$$\tilde{x}_1 = \frac{221 + 222}{2} = 221.5.$$

For aging, the median is

$$\tilde{x}_2 = \frac{209 + 211}{2} = 210.$$

Thus

$$\tilde{x}_{\text{no aging}} = 221.5, \quad \tilde{x}_{\text{aging}} = 210.$$

In both groups, the mean and median are very close:

$$222.1 \text{ vs. } 221.5, \quad 209.9 \text{ vs. } 210.$$

This suggests that neither group is strongly skewed.

1.4

In a study conducted by the Department of Mechanical Engineering at Virginia Tech, the steel rods supplied by two different companies were compared. Ten sample springs were made out of the steel rods supplied by each company, and a measure of flexibility was recorded for each. The data are as follows:

Company A:

$$9.3, 8.8, 6.8, 8.7, 8.5, 6.7, 8.0, 6.5, 9.2, 7.0$$

Company B:

$$11.0, 9.8, 9.9, 10.2, 10.1, 9.7, 11.0, 11.1, 10.2, 9.6$$

- Find the sample mean and median for the data for the two companies.
- Plot the data for the two companies on the same line and give your impression regarding any apparent differences between the two companies.

Solution

(a) For Company A:

$$\bar{x}_A = \frac{79.5}{10} = 7.95.$$

Ordered data:

$$6.5, 6.7, 6.8, 7.0, 8.0, 8.5, 8.7, 8.8, 9.2, 9.3.$$

Median:

$$\tilde{x}_A = \frac{8.0 + 8.5}{2} = 8.25.$$

For Company B:

$$\bar{x}_B = \frac{102.6}{10} = 10.26.$$

Ordered data:

$$9.6, 9.7, 9.8, 9.9, 10.1, 10.2, 10.2, 11.0, 11.0, 11.1.$$

Median:

$$\tilde{x}_B = \frac{10.1 + 10.2}{2} = 10.15.$$

Therefore

$$\bar{x}_A = 7.95, \quad \tilde{x}_A = 8.25, \quad \bar{x}_B = 10.26, \quad \tilde{x}_B = 10.15.$$

- (b) Company B's observations are all shifted to the right of Company A's observations. So Company B tends to produce rods with larger flexibility measurements than Company A. There is only slight overlap, so the difference appears substantial.

Company B appears to produce consistently higher flexibility values than Company A.

1.5

Twenty adult males between the ages of 30 and 40 participated in a study to evaluate the effect of a specific health regimen involving diet and exercise on the blood cholesterol. Ten were randomly selected to be a control group, and ten others were assigned to take part in the regimen as the treatment group for a period of 6 months. The following data show the reduction in cholesterol experienced by the 20 subjects:

Control group:

7, 3, -4, 14, 2, 5, 22, -7, 9, 5

Treatment group:

-6, 5, 9, 4, 4, 12, 37, 5, 3, 3

- (a) Do a dot plot of the data for both groups on the same graph.
- (b) Compute the mean, median, and 10% trimmed mean for both groups.
- (c) Explain why the difference in means suggests one conclusion about the effect of the regimen, while the difference in medians or trimmed means suggests a different conclusion.

Solution

- (a) Ordered data:

Control:

-7, -4, 2, 3, 5, 5, 7, 9, 14, 22

Treatment:

-6, 3, 3, 4, 4, 5, 5, 9, 12, 37

A common dot plot would show that the treatment group contains one very large value, 37.

- (b) For the control group:

$$\bar{x}_C = \frac{56}{10} = 5.6, \quad \tilde{x}_C = \frac{5 + 5}{2} = 5.$$

For a 10% trimmed mean, remove the smallest and largest values:

-7, 22.

Remaining sum:

$$56 - (-7) - 22 = 41,$$

so

$$\bar{x}_{C,0.10} = \frac{41}{8} = 5.125.$$

For the treatment group:

$$\bar{x}_T = \frac{76}{10} = 7.6, \quad \tilde{x}_T = \frac{4 + 5}{2} = 4.5.$$

Trimmed mean: remove

$$-6, 37.$$

Remaining sum:

$$76 - (-6) - 37 = 45,$$

so

$$\bar{x}_{T,0.10} = \frac{45}{8} = 5.625.$$

Therefore

$$\bar{x}_C = 5.6, \quad \tilde{x}_C = 5, \quad \bar{x}_{C,0.10} = 5.125$$

$$\bar{x}_T = 7.6, \quad \tilde{x}_T = 4.5, \quad \bar{x}_{T,0.10} = 5.625$$

- (c) The treatment mean is higher than the control mean:

$$7.6 > 5.6,$$

which suggests the treatment may be more effective.

However, this is driven heavily by the unusually large observation 37. The medians and trimmed means are much closer:

$$4.5 \text{ vs. } 5, \quad 5.625 \text{ vs. } 5.125.$$

These more resistant measures suggest that the typical treatment effect is not dramatically different from the control.

The ordinary mean is strongly influenced by the outlier 37, whereas the median and trimmed mean are

1.6

The tensile strength of silicone rubber is thought to be a function of curing temperature. A study was carried out using samples of 12 specimens of the rubber prepared using curing temperatures of 20°C and 45°C . The values below show the tensile strength values in megapascals.

20°C :

$$2.07, 2.14, 2.22, 2.03, 2.21, 2.03, 2.05, 2.18, 2.09, 2.14, 2.11, 2.02$$

45°C :

$$2.52, 2.15, 2.49, 2.03, 2.37, 2.05, 1.99, 2.42, 2.08, 2.42, 2.29, 2.01$$

- Show a dot plot of the data with both low and high temperature tensile strength values.
- Compute sample mean tensile strength for both samples.
- Does it appear as if curing temperature has an influence on tensile strength, based on the plot? Comment further.
- Does anything else appear to be influenced by an increase in curing temperature? Explain.

Solution

(a) Ordered data:

20°C:

2.02, 2.03, 2.03, 2.05, 2.07, 2.09, 2.11, 2.14, 2.14, 2.18, 2.21, 2.22

45°C:

1.99, 2.01, 2.03, 2.05, 2.08, 2.15, 2.29, 2.37, 2.42, 2.42, 2.49, 2.52

A combined dot plot would show the 45°C values generally shifted upward, but also much more spread out.

(b) Sample means:

For 20°C,

$$\bar{x}_{20} = \frac{25.29}{12} = 2.1075.$$

For 45°C,

$$\bar{x}_{45} = \frac{26.82}{12} = 2.235.$$

Hence

$$\bar{x}_{20} = 2.1075, \quad \bar{x}_{45} = 2.235.$$

(c) Yes. The mean at 45°C is larger, and the dot plot would show more high values at the higher temperature. This suggests that increasing curing temperature tends to increase tensile strength.

Higher curing temperature appears to increase tensile strength on average.

(d) Yes. The variability also appears to increase at the higher temperature. The 20°C values are tightly clustered around 2.1, whereas the 45°C values range from about 1.99 to 2.52, showing noticeably more spread.

An increase in curing temperature appears to increase both the mean tensile strength and the variability.

Exercises 1.7–1.12 with Solutions**1.7**

Consider the drying time data for Exercise 1.1 on page 13. Compute the sample variance and sample standard deviation.

Solution

The data from Exercise 1.1 are

3.4, 2.5, 4.8, 2.9, 3.6, 2.8, 3.3, 5.6, 3.7, 2.8, 4.4, 4.0, 5.2, 3.0, 4.8.

From Exercise 1.1, the sample mean is

$$\bar{x} = 3.7867.$$

Using the sample variance formula,

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2, \quad n = 15,$$

we obtain

$$s^2 = 0.9427.$$

Therefore, the sample standard deviation is

$$s = \sqrt{0.9427} = 0.9709.$$

$s^2 \approx 0.9427, \quad s \approx 0.9709.$

1.8

Compute the sample variance and sample standard deviation for the water absorbency data of Exercise 1.2 on page 13.

Solution

The data from Exercise 1.2 are

$$\begin{aligned} &18.71, 21.41, 20.72, 21.81, 19.29, 22.43, 20.17, \\ &23.71, 19.44, 20.50, 18.92, 20.33, 23.00, 22.85, \\ &19.25, 21.77, 22.11, 19.77, 18.04, 21.12. \end{aligned}$$

From Exercise 1.2, the sample mean is

$$\bar{x} = 20.7675.$$

Using

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2, \quad n = 20,$$

we get

$$s^2 = 2.5329.$$

Hence the sample standard deviation is

$$s = \sqrt{2.5329} = 1.5915.$$

$s^2 \approx 2.5329, \quad s \approx 1.5915.$

1.9

Exercise 1.3 on page 13 showed tensile strength data for two samples, one in which specimens were exposed to an aging process and one in which there was no aging of the specimens.

- (a) Calculate the sample variance as well as standard deviation in tensile strength for both samples.
- (b) Does there appear to be any evidence that aging affects the variability in tensile strength? (See also the plot for Exercise 1.3 on page 13.)

Solution

The data from Exercise 1.3 are:

No aging:

227, 222, 218, 217, 225, 218, 216, 229, 228, 221

Aging:

219, 214, 215, 211, 209, 218, 203, 204, 201, 205

From Exercise 1.3, the sample means are

$$\bar{x}_{\text{no aging}} = 222.1, \quad \bar{x}_{\text{aging}} = 209.9.$$

(a) Variance and standard deviation. For the no-aging sample:

$$s_1^2 = \frac{1}{9} \sum (x_i - \bar{x}_1)^2 = 23.6556,$$

so

$$s_1 = \sqrt{23.6556} = 4.8637.$$

For the aging sample:

$$s_2^2 = \frac{1}{9} \sum (x_i - \bar{x}_2)^2 = 42.1000,$$

so

$$s_2 = \sqrt{42.1000} = 6.4885.$$

Thus

$$s_{\text{no aging}}^2 \approx 23.6556, \quad s_{\text{no aging}} \approx 4.8637,$$

$$s_{\text{aging}}^2 = 42.1000, \quad s_{\text{aging}} \approx 6.4885.$$

(b) Comment on variability. Yes. The aged sample has a noticeably larger variance and standard deviation:

$$42.10 > 23.66, \quad 6.49 > 4.86.$$

So the data suggest that aging increases not only the reduction in mean tensile strength, but also the variability in tensile strength.

There is evidence that aging increases the variability in tensile strength.

1.10

For the data of Exercise 1.4 on page 13, compute both the mean and variance in “flexibility” for both company A and company B. Does there appear to be a difference in flexibility between company A and company B?

Solution

The data from Exercise 1.4 are:

Company A:

9.3, 8.8, 6.8, 8.7, 8.5, 6.7, 8.0, 6.5, 9.2, 7.0

Company B:

11.0, 9.8, 9.9, 10.2, 10.1, 9.7, 11.0, 11.1, 10.2, 9.6

For Company A:

$$\bar{x}_A = \frac{79.5}{10} = 7.95,$$

and

$$s_A^2 = 1.2072.$$

For Company B:

$$\bar{x}_B = \frac{102.6}{10} = 10.26,$$

and

$$s_B^2 = 0.3249.$$

Therefore,

$$\bar{x}_A = 7.95, \quad s_A^2 \approx 1.2072,$$

$$\bar{x}_B = 10.26, \quad s_B^2 \approx 0.3249.$$

Since

$$10.26 > 7.95,$$

Company B has a substantially larger average flexibility measure than Company A. Also, Company B shows less variability.

Yes, there appears to be a clear difference: Company B has higher average flexibility and smaller variance.

1.11

Consider the data in Exercise 1.5 on page 13. Compute the sample variance and the sample standard deviation for both control and treatment groups.

Solution

The data from Exercise 1.5 are:

Control group:

$$7, 3, -4, 14, 2, 5, 22, -7, 9, 5$$

Treatment group:

$$-6, 5, 9, 4, 4, 12, 37, 5, 3, 3$$

From Exercise 1.5, the sample means are

$$\bar{x}_C = 5.6, \quad \bar{x}_T = 7.6.$$

For the control group,

$$s_C^2 = \frac{1}{9} \sum (x_i - \bar{x}_C)^2 = 69.3778,$$

so

$$s_C = \sqrt{69.3778} = 8.3293.$$

For the treatment group,

$$s_T^2 = \frac{1}{9} \sum (x_i - \bar{x}_T)^2 = 128.0444,$$

so

$$s_T = \sqrt{128.0444} = 11.3157.$$

Hence

$$s_C^2 \approx 69.3778, \quad s_C \approx 8.3293,$$

$$s_T^2 \approx 128.0444, \quad s_T \approx 11.3157.$$

The treatment group has much larger variability, mainly because of the large observation 37.

1.12

For Exercise 1.6 on page 13, compute the sample standard deviation in tensile strength for the samples separately for the two temperatures. Does it appear as if an increase in temperature influences the variability in tensile strength? Explain.

Solution

The data from Exercise 1.6 are:

20°C:

2.07, 2.14, 2.22, 2.03, 2.21, 2.03, 2.05, 2.18, 2.09, 2.14, 2.11, 2.02

45°C:

2.52, 2.15, 2.49, 2.03, 2.37, 2.05, 1.99, 2.42, 2.08, 2.42, 2.29, 2.01

For the 20°C sample:

$$s_{20}^2 = 0.00502045, \quad s_{20} = \sqrt{0.00502045} = 0.07086.$$

For the 45°C sample:

$$s_{45}^2 = 0.04128182, \quad s_{45} = \sqrt{0.04128182} = 0.20318.$$

Therefore,

$$s_{20} \approx 0.0709, \quad s_{45} \approx 0.2032.$$

Since

$$0.2032 > 0.0709,$$

the higher-temperature sample has much greater spread. So the data suggest that increasing temperature increases the variability in tensile strength.

Yes, an increase in temperature appears to increase the variability in tensile strength.

1.13

A manufacturer of electronic components is interested in determining the lifetime of a certain type of battery. A sample, in hours of life, is as follows:

123, 116, 122, 110, 175, 126, 125, 111, 118, 117.

- Find the sample mean and median.
- What feature in this data set is responsible for the substantial difference between the two?

Solution

The sample size is $n = 10$.

$$\bar{x} = \frac{123 + 116 + 122 + 110 + 175 + 126 + 125 + 111 + 118 + 117}{10} = \frac{1243}{10} = 124.3.$$

To find the median, first sort the data:

$$110, 111, 116, 117, 118, 122, 123, 125, 126, 175.$$

Since $n = 10$ is even, the median is the average of the 5th and 6th observations:

$$\tilde{x} = \frac{118 + 122}{2} = 120.$$

$$\boxed{\bar{x} = 124.3, \quad \tilde{x} = 120}$$

For part (b), the value 175 is much larger than the rest and acts as an outlier. It pulls the mean upward much more than it affects the median.

The substantial difference is caused by the high outlier 175.

1.14

A tire manufacturer wants to determine the inner diameter of a certain grade of tire. Ideally, the diameter would be 570 mm. The data are as follows:

$$572, 572, 573, 568, 569, 575, 565, 570.$$

- (a) Find the sample mean and median.
- (b) Find the sample variance, standard deviation, and range.
- (c) Using the calculated statistics in parts (a) and (b), can you comment on the quality of the tires?

Solution

Here $n = 8$.

$$\bar{x} = \frac{572 + 572 + 573 + 568 + 569 + 575 + 565 + 570}{8} = \frac{4564}{8} = 570.5.$$

Sort the data:

$$565, 568, 569, 570, 572, 572, 573, 575.$$

Median:

$$\tilde{x} = \frac{570 + 572}{2} = 571.$$

So,

$$\boxed{\bar{x} = 570.5, \quad \tilde{x} = 571}$$

Now compute the sample variance:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

Using $\bar{x} = 570.5$, the deviations are:

$$1.5, 1.5, 2.5, -2.5, -1.5, 4.5, -5.5, -0.5.$$

Their squares are:

$$2.25, 2.25, 6.25, 6.25, 2.25, 20.25, 30.25, 0.25.$$

Thus,

$$\sum (x_i - \bar{x})^2 = 70.$$

Hence

$$s^2 = \frac{70}{7} = 10, \quad s = \sqrt{10} \approx 3.162.$$

The range is

$$\text{Range} = 575 - 565 = 10.$$

Therefore,

$$\boxed{s^2 = 10, \quad s \approx 3.162, \quad \text{Range} = 10}$$

For part (c), the mean and median are both very close to the target value 570, and the variability is not large. This suggests that the tire diameters are centered close to the desired specification and appear reasonably consistent.

The tire quality appears satisfactory: the diameters are close to 570 mm with moderate variability.

1.15

Five independent coin tosses result in

$$HHHHH.$$

It turns out that if the coin is fair the probability of this outcome is

$$(1/2)^5 = 0.03125.$$

Does this produce strong evidence that the coin is not fair? Comment and use the concept of P -value discussed in Section 1.1.

Solution

Under the null hypothesis

$$H_0 : \text{the coin is fair,}$$

the probability of observing five heads in five tosses is

$$P(HHHHH) = \left(\frac{1}{2}\right)^5 = 0.03125.$$

If we define the evidence against fairness in the direction of “too many heads,” then the observed result is very unusual. Its one-sided P -value is

$$P(X = 5) = 0.03125,$$

where $X \sim \text{Binomial}(5, 1/2)$.

If instead we use a two-sided viewpoint (extreme in either direction: all heads or all tails), then

$$P(X = 5 \text{ or } X = 0) = 2 \left(\frac{1}{2}\right)^5 = 0.0625.$$

Thus the evidence is suggestive, but not overwhelmingly strong. In many settings:

- one-sided: $0.03125 < 0.05$, so one might reject at the 5% level;
- two-sided: $0.0625 > 0.05$, so one would not reject at the 5% level.

Five heads in a row is suspicious, but by itself it is not very strong evidence that the coin is unfair.

1.16

Show that the n pieces of information in

$$\sum_{i=1}^n (x_i - \bar{x})^2$$

are not independent; that is, show that

$$\sum_{i=1}^n (x_i - \bar{x}) = 0.$$

Solution

Recall that

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

Then

$$\sum_{i=1}^n (x_i - \bar{x}) = \sum_{i=1}^n x_i - \sum_{i=1}^n \bar{x}.$$

Since $\bar{x}$ is constant with respect to the index i ,

$$\sum_{i=1}^n \bar{x} = n\bar{x}.$$

Therefore,

$$\sum_{i=1}^n (x_i - \bar{x}) = \sum_{i=1}^n x_i - n\bar{x}.$$

But

$$n\bar{x} = n \left(\frac{1}{n} \sum_{i=1}^n x_i \right) = \sum_{i=1}^n x_i.$$

Hence,

$$\sum_{i=1}^n (x_i - \bar{x}) = 0.$$

$$\sum_{i=1}^n (x_i - \bar{x}) = 0$$

This shows the deviations from the sample mean are linearly dependent, so the n terms do not provide n independent pieces of information.

1.17

A study of the effects of smoking on sleep patterns is conducted. The measure observed is the time, in minutes, that it takes to fall asleep. These data are obtained:

Smokers:

69.3, 56.0, 22.1, 47.6, 53.2, 48.1, 52.7, 34.4, 60.2, 43.8, 23.2, 13.8

Nonsmokers:

28.6, 25.1, 26.4, 34.9, 29.8, 28.4, 38.5, 30.2, 30.6, 31.8, 41.6, 21.1, 36.0, 37.9, 13.9

- Find the sample mean for each group.
- Find the sample standard deviation for each group.
- Make a dot plot of the data sets A and B on the same line.
- Comment on what kind of impact smoking appears to have on the time required to fall asleep.

Solution

(a) Sample means. For smokers ($n_1 = 12$):

$$\bar{x}_1 = \frac{1}{12} \sum x_i = 43.7.$$

For nonsmokers ($n_2 = 15$):

$$\bar{x}_2 = \frac{1}{15} \sum x_i = 30.32.$$

Thus,

$\bar{x}_{\text{smokers}} = 43.7,$	$\bar{x}_{\text{nonsmokers}} = 30.32$
------------------------------------	---------------------------------------

(b) Sample standard deviations. Using

$$s = \sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2},$$

we obtain:

$$s_{\text{smokers}} \approx 16.93, \quad s_{\text{nonsmokers}} \approx 7.13.$$

So,

$s_{\text{smokers}} \approx 16.93,$	$s_{\text{nonsmokers}} \approx 7.13$
-------------------------------------	--------------------------------------

(c) Dot plot. A full graphical dot plot can be added later in TikZ/PGFPlots, but the ordered data already show the pattern clearly.

Ordered smokers:

13.8, 22.1, 23.2, 34.4, 43.8, 47.6, 48.1, 52.7, 53.2, 56.0, 60.2, 69.3

Ordered nonsmokers:

13.9, 21.1, 25.1, 26.4, 28.4, 28.6, 29.8, 30.2, 30.6, 31.8, 34.9, 36.0, 37.9, 38.5, 41.6

(d) **Comment.** The smokers have:

- a much larger mean time to fall asleep;
- a much larger spread in times.

This suggests that smoking is associated with taking longer to fall asleep, and the sleep onset times among smokers are also more variable.

Smoking appears to increase the time required to fall asleep and also increases variability.

1.18

The following scores represent the final examination grades for an elementary statistics course:

23	60	79	32	57	74	52	70	82
36	80	77	81	95	41	65	92	85
55	76	52	10	64	75	78	25	80
98	81	67	41	71	83	54	64	72
88	62	74	43	60	78	89	76	84
48	84	90	15	79	34	67	17	82
69	74	63	80	85	61			

- Construct a stem-and-leaf plot for the examination grades in which the stems are 1, 2, 3, ..., 9.
- Construct a relative frequency histogram, draw an estimate of the graph of the distribution, and discuss the skewness of the distribution.
- Compute the sample mean, sample median, and sample standard deviation.

Solution

(a) **Stem-and-leaf plot.** Using the tens digit as the stem and the ones digit as the leaf:

1	0 5 7
2	3 5
3	2 4 6
4	1 1 3 8
5	2 2 4 5 7
6	0 0 1 2 3 4 4 5 7 7 9
7	0 1 2 4 4 4 5 6 6 7 8 8 9 9
8	0 0 0 1 1 2 2 3 4 4 5 5 8 9
9	0 2 5 8

(b) **Distribution and skewness.** Most scores cluster in the 60s, 70s, and 80s, with a smaller number of low scores in the teens, 20s, and 30s. This suggests a left-skewed (negatively skewed) distribution: there is a longer tail toward the lower scores.

(c) **Sample mean, median, and standard deviation.** There are $n = 60$ observations. The computed statistics are:

$$\bar{x} = 65.4833, \quad \tilde{x} = 71.5, \quad s \approx 21.1335.$$

Thus,

$$\bar{x} \approx 65.48, \quad \tilde{x} = 71.5, \quad s \approx 21.13$$

Since $\bar{x} < \tilde{x}$, this also supports the conclusion that the distribution is negatively skewed.

1.21

The lengths of power failures, in minutes, are recorded in the following table.

22	18	135	15	90	78	69	98	102
83	55	28	121	120	13	22	124	112
70	66	74	89	103	24	21	112	21
40	98	87	132	115	21	28	43	37
50	96	118	158	74	78	83	93	95

- Find the sample mean and sample median of the power-failure times.
- Find the sample standard deviation of the power-failure times.

Solution

There are $n = 45$ observations.

(a) Mean and median. The sample mean is

$$\bar{x} = \frac{1}{45} \sum x_i \approx 74.0222.$$

The sample median is

$$\tilde{x} = 78.$$

Hence,

$$\boxed{\bar{x} \approx 74.02, \quad \tilde{x} = 78}$$

(b) Sample standard deviation. Using the sample standard deviation formula,

$$s = \sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2},$$

we obtain

$$\boxed{s \approx 39.26}$$

The data show substantial spread, with some very short outages and some very long outages.

1.22

The following data are the measures of the diameters of 36 rivet heads in 1/100 of an inch.

6.72	6.77	6.82	6.70	6.78	6.70	6.62	6.75
6.66	6.66	6.64	6.76	6.73	6.80	6.72	6.76
6.76	6.68	6.66	6.62	6.72	6.76	6.70	6.78
6.76	6.67	6.70	6.72	6.74	6.81	6.79	6.78
6.66	6.76	6.76	6.72				

- Compute the sample mean and sample standard deviation.
- Construct a relative frequency histogram of the data.
- Comment on whether or not there is any clear indication that the sample came from a population that has a bell-shaped distribution.

Solution

(a) **Mean and standard deviation.** For $n = 36$,

$$\bar{x} \approx 6.7261, \quad s \approx 0.0536.$$

Thus,

$$\boxed{\bar{x} \approx 6.7261, \quad s \approx 0.0536}$$

(b) **Relative frequency histogram.** The values are tightly concentrated between approximately 6.62 and 6.82. A histogram would show a unimodal cluster centered near 6.73.

(c) **Comment.** There is no obvious strong skewness or extreme outlier. The data appear roughly symmetric and concentrated around one center, so a bell-shaped population is plausible.

The data are reasonably consistent with a bell-shaped distribution.

1.24

The following are historical data on staff salaries (dollars per pupil) for 30 schools sampled in the eastern part of the United States in the early 1970s.

3.79	2.99	2.77	2.91	3.10	1.84	2.52	3.22
2.45	2.14	2.67	2.52	2.71	2.75	3.57	3.85
3.36	2.05	2.89	2.83	3.13	2.44	2.10	3.71
3.14	3.54	2.37	2.68	3.51	3.37		

(a) Compute the sample mean and sample standard deviation.

(b) Construct a relative frequency histogram of the data.

(c) Construct a stem-and-leaf display of the data.

Solution

(a) **Mean and standard deviation.** For $n = 30$,

$$\bar{x} \approx 2.8973, \quad s \approx 0.5415.$$

So,

$$\boxed{\bar{x} \approx 2.8973, \quad s \approx 0.5415}$$

(b) **Histogram.** A histogram would show most observations between 2.4 and 3.6, with a low value at 1.84. The data appear moderately spread and slightly asymmetric.

(c) **Stem-and-leaf display.** Using the integer part and first decimal digit as the stem, and the second decimal digit as the leaf:

1.8	4
2.0	5
2.1	0 4
2.3	7
2.4	4 5
2.5	2 2
2.6	7 8
2.7	1 5 7
2.8	3 9
2.9	1 9
3.1	0 3 4
3.2	2
3.3	6 7
3.5	1 4 7
3.7	1 9
3.8	5

1.25

The following data set is related to that in Exercise 1.24. It gives the percentages of the families that are in the upper income level, for the same individual schools in the same order as in Exercise 1.24.

72.2	31.9	26.5	29.1	27.3	8.6	22.3	26.5
20.4	12.8	25.1	19.2	24.1	58.2	68.1	89.2
55.1	9.4	14.5	13.9	20.7	17.9	8.5	55.4
38.1	54.2	21.5	26.2	59.1	43.3		

- Calculate the sample mean.
- Calculate the sample median.
- Construct a relative frequency histogram of the data.
- Compute the 10% trimmed mean. Compare with the results in (a) and (b) and comment.

Solution

There are $n = 30$ observations.

(a) Sample mean.

$$\bar{x} = \frac{1}{30} \sum x_i = 33.31.$$

$$\boxed{\bar{x} = 33.31}$$

(b) Sample median. After sorting the data, the median is the average of the 15th and 16th values:

$$\tilde{x} = 26.35.$$

$$\boxed{\tilde{x} = 26.35}$$

(c) Histogram. The data are spread widely and include several large values such as 68.1, 72.2, 89.2. A histogram would likely be right-skewed.

(d) Ten percent trimmed mean. Since 10% of 30 is 3, remove the three smallest and three largest observations.

Sorted extremes:

lowest: 8.5, 8.6, 9.4, highest: 68.1, 72.2, 89.2.

The mean of the remaining 24 observations is

$$\bar{x}_t = 30.9708.$$

So,

$$\boxed{\bar{x}_t \approx 30.97}$$

Comparison:

$$\bar{x} = 33.31, \quad \tilde{x} = 26.35, \quad \bar{x}_t \approx 30.97.$$

The trimmed mean lies between the ordinary mean and the median, which is typical when the data are right-skewed. The large upper-end values inflate the ordinary mean.

The data are right-skewed; trimming reduces the effect of the large upper-tail observations.

1.26

Suppose it is of interest to use the data sets in Exercises 1.24 and 1.25 to derive a model that would predict staff salaries as a function of percentage of families in a high income level for current school systems. Comment on any disadvantage in carrying out this type of analysis.

Solution

A major disadvantage is that the data are historical and come from the early 1970s. Therefore, a model built from these data may not generalize well to current school systems because:

- salary structures have changed substantially over time;
- inflation and regional cost-of-living differences matter;
- education funding rules and labor markets may have changed;
- the relationship may be confounded by omitted variables such as school size, district policies, unionization, urban/rural location, and public funding formulas.

Also, these data are observational, not experimental. So even if a strong relationship exists, one should not interpret it as causal.

The main disadvantages are outdated data, omitted-variable bias, and the non-causal nature of observational

1.28

Many manufacturing companies in the United States and abroad use molded parts as components of a process. Shrinkage is often a major problem. Thus, a molded die for a part is built larger than nominal size to allow for part shrinkage. In an injection molding study it is known that the shrinkage is influenced by many factors, among which are the injection velocity in ft/sec and mold temperature in °C. The following two data sets show the results of a designed experiment in which injection velocity was held at two levels (low and high) and mold temperature was held constant at a low level. The shrinkage is measured in $\text{cm} \times 10^4$.

Shrinkage values at low injection velocity:

72.68, 72.62, 72.58, 72.48, 73.07, 72.55, 72.42, 72.84, 72.58, 72.92

Shrinkage values at high injection velocity:

71.62, 71.68, 71.74, 71.48, 71.55, 71.52, 71.71, 71.56, 71.70, 71.50

- Construct a dot plot of both data sets on the same graph. Indicate on the plot both shrinkage means, that for low injection velocity and high injection velocity.
- Based on the graphical results in (a), using the location of the two means and your sense of variability, what do you conclude regarding the effect of injection velocity on shrinkage at low mold temperature?

Solution

(a) Sample means. Let group L denote low injection velocity and group H denote high injection velocity.

For low injection velocity,

$$\bar{x}_L = \frac{72.68 + 72.62 + 72.58 + 72.48 + 73.07 + 72.55 + 72.42 + 72.84 + 72.58 + 72.92}{10} = \frac{726.74}{10} = 72.674.$$

For high injection velocity,

$$\bar{x}_H = \frac{71.62 + 71.68 + 71.74 + 71.48 + 71.55 + 71.52 + 71.71 + 71.56 + 71.70 + 71.50}{10} = \frac{716.06}{10} = 71.606.$$

Thus,

$$\boxed{\bar{x}_L = 72.674, \quad \bar{x}_H = 71.606}$$

The difference in means is

$$\bar{x}_L - \bar{x}_H = 72.674 - 71.606 = 1.068.$$

Variability within each group. To judge the separation, it is helpful to compute the sample standard deviations.

For the low-velocity group,

$$s_L = \sqrt{\frac{1}{9} \sum_{i=1}^{10} (x_i - \bar{x}_L)^2} \approx 0.214.$$

For the high-velocity group,

$$s_H = \sqrt{\frac{1}{9} \sum_{i=1}^{10} (x_i - \bar{x}_H)^2} \approx 0.098.$$

So,

$$\boxed{s_L \approx 0.214, \quad s_H \approx 0.098}$$

Dot plot interpretation. A dot plot would place the low-velocity observations roughly between 72.42 and 73.07, and the high-velocity observations roughly between 71.48 and 71.74. The two groups are clearly separated, with almost no overlap. Their means, 72.674 and 71.606, would appear far apart relative to the spread inside each group.

(b) Conclusion about the effect of injection velocity at low mold temperature. At low mold temperature, the mean shrinkage for low injection velocity is larger than the mean shrinkage for high injection velocity by about

$$1.068.$$

This difference is very large compared with the within-group variability.

Therefore, based on the graphical separation of the two groups and the difference in means, we conclude that injection velocity has a strong effect on shrinkage when mold temperature is held low. In particular, increasing injection velocity appears to *decrease* shrinkage.

At low mold temperature, higher injection velocity is associated with substantially smaller shrinkage.

1.30

Below are the lifetimes, in hours, of fifty 40-watt, 110-volt internally frosted incandescent lamps, taken from forced life tests:

919	1196	785	1126	936	918
1156	920	948	1067	1092	1162
1170	929	950	905	972	1035
1045	855	1195	1195	1340	1122
938	970	1237	956	1102	1157
978	832	1009	1157	1151	1009
765	958	902	1022	1333	811
1217	1085	896	958	1311	1037
702	923				

Construct a box plot for these data.

Solution

To construct the box plot, compute the five-number summary.

$$\min = 702, \quad \max = 1340.$$

The median is

$$Q_2 = 1009.$$

The first and third quartiles are approximately

$$Q_1 = 924.5, \quad Q_3 = 1154.75.$$

Thus the five-number summary is

$$(702, 924.5, 1009, 1154.75, 1340).$$

The interquartile range is

$$IQR = Q_3 - Q_1 = 1154.75 - 924.5 = 230.25.$$

Outlier fences:

$$Q_1 - 1.5(IQR) = 924.5 - 1.5(230.25) = 579.125,$$

$$Q_3 + 1.5(IQR) = 1154.75 + 1.5(230.25) = 1500.125.$$

Since both 702 and 1340 lie within the fences, there are no outliers by the $1.5 \times IQR$ rule. So the box plot should have:

- left whisker at 702,
- box from 924.5 to 1154.75,
- median line at 1009,
- right whisker at 1340,
- no plotted outliers.

1.32

Use the results of Exercises 1.28 and 1.31 to create a plot that illustrates the interaction evident from the data. Use the plot in Figure 1.3 in Example 1.3 as a guide. Could the type of information found in Exercises 1.28 and 1.31 have been found in an observational study in which there was no control on injection velocity and mold temperature by the analyst? Explain why or why not.

Solution

Interaction plot. An interaction plot uses mold temperature on the horizontal axis and mean shrinkage on the vertical axis, with separate lines for low and high injection velocity.

From Exercise 1.28 (low mold temperature):

$$\text{Low velocity mean} = 72.674, \quad \text{High velocity mean} = 71.606.$$

From Exercise 1.31 (high mold temperature):

$$\text{Low velocity mean} = 76.09, \quad \text{High velocity mean} = 93.168.$$

So the plot is based on these four points:

$$(\text{Low temp}, 72.674), \quad (\text{High temp}, 76.09)$$

for the low-velocity line, and

$$(\text{Low temp}, 71.606), \quad (\text{High temp}, 93.168)$$

for the high-velocity line.

Because these two lines are not parallel and, in fact, show very different changes across temperature, the interaction plot clearly displays interaction.

Interpretation of the interaction plot. For low injection velocity, raising mold temperature increases shrinkage moderately:

$$76.09 - 72.674 = 3.416.$$

For high injection velocity, raising mold temperature increases shrinkage much more:

$$93.168 - 71.606 = 21.562.$$

Likewise, the effect of increasing injection velocity changes direction depending on mold temperature:

- at low mold temperature: shrinkage decreases by 1.068;
- at high mold temperature: shrinkage increases by 17.078.

This nonparallel pattern is the hallmark of interaction.

The interaction plot clearly shows a strong interaction between injection velocity and mold temperature.

Could this have been found in an observational study? Not nearly as reliably. In an observational study, the analyst does not control injection velocity and mold temperature. As a result:

- the two factors may be associated with other process conditions;
- confounding variables may distort the apparent relationship;
- some combinations of factor levels may occur rarely or not at all;
- it is much harder to separate the effect of velocity, temperature, and their interaction.

In contrast, a designed experiment deliberately controls the factor levels and compares responses under structured conditions. That design makes the interaction visible and interpretable.

The interaction could be obscured in an observational study because of lack of control and possible confounding.

Chapter 2: Exercises with Solutions

2.1

List the elements of each of the following sample spaces:

- (a) the set of integers between 1 and 50 divisible by 8;
- (b) the set $S = \{x \mid x^2 + 4x - 5 = 0\}$;
- (c) the set of outcomes when a coin is tossed until a tail or three heads appear;
- (d) the set $S = \{x \mid x \text{ is a continent}\}$;
- (e) the set $S = \{x \mid 2x - 4 \geq 0 \text{ and } x < 1\}$.

Solution

- (a) Multiples of 8 between 1 and 50:

$$S = \{8, 16, 24, 32, 40, 48\}.$$

- (b) Solve

$$x^2 + 4x - 5 = 0 = (x + 5)(x - 1).$$

Hence

$$S = \{-5, 1\}.$$

- (c) Toss until the first tail appears or until three heads have appeared:

$$S = \{T, HT, HHT, HHH\}.$$

- (d) Continents:

$$S = \{\text{Africa, Antarctica, Asia, Europe, North America, Australia, South America}\}.$$

- (e) Conditions:

$$2x - 4 \geq 0 \implies x \geq 2, \quad x < 1.$$

These cannot both hold, so

$$S = \emptyset.$$

2.2

Use the rule method to describe the sample space S consisting of all points in the first quadrant inside a circle of radius 3 with center at the origin.

Solution

Points in the first quadrant satisfy $x \geq 0$, $y \geq 0$, and points inside the circle of radius 3 satisfy $x^2 + y^2 \leq 9$. Therefore,

$$S = \{(x, y) \mid x \geq 0, y \geq 0, x^2 + y^2 \leq 9\}.$$

2.3

Which of the following events are equal?

$$(a) A = \{1, 3\}; \quad (b) B = \{x \mid x \text{ is a number on a die}\};$$

$$(c) C = \{x \mid x^2 - 4x + 3 = 0\}; \quad (d) D = \{x \mid x \text{ is the number of heads when six coins are tossed}\}.$$

Solution

We determine each set:

$$A = \{1, 3\}.$$

For a die,

$$B = \{1, 2, 3, 4, 5, 6\}.$$

Solve

$$x^2 - 4x + 3 = 0 = (x - 1)(x - 3),$$

so

$$C = \{1, 3\}.$$

The number of heads in six tosses can be

$$D = \{0, 1, 2, 3, 4, 5, 6\}.$$

Hence

$$\boxed{A = C}$$

and no other pairs are equal.

2.4

An experiment involves tossing a pair of dice, one green and one red, and recording the numbers that come up. If x equals the outcome on the green die and y the outcome on the red die, describe the sample space S

- (a) by listing the elements (x, y) ;
- (b) by using the rule method.

Solution

- (a) The sample space is

$$\begin{aligned} S = \{ & (1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), \\ & (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), \\ & (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), \\ & (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), \\ & (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), \\ & (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6) \}. \end{aligned}$$

- (b) In rule form,

$$S = \{(x, y) \mid x \in \{1, 2, 3, 4, 5, 6\}, y \in \{1, 2, 3, 4, 5, 6\}\}.$$

2.5

An experiment consists of tossing a die and then flipping a coin once if the number on the die is even. If the number on the die is odd, the coin is flipped twice. Using the notation $4H$, for example, to denote the outcome that the die comes up 4 and then the coin comes up heads, and $3HT$ to denote the outcome that the die comes up 3 followed by a head and then a tail on the coin, construct a tree diagram to show the 18 elements of the sample space S .

Solution

If the die is even (2, 4, 6), there is one coin flip: H or T . If the die is odd (1, 3, 5), there are two coin flips: HH, HT, TH, TT .

Thus

$$\begin{aligned} S = \{ & 1HH, 1HT, 1TH, 1TT, \\ & 2H, 2T, \\ & 3HH, 3HT, 3TH, 3TT, \\ & 4H, 4T, \\ & 5HH, 5HT, 5TH, 5TT, \\ & 6H, 6T \}. \end{aligned}$$

There are

$$4 + 2 + 4 + 2 + 4 + 2 = 18$$

outcomes.

2.6

Two jurors are selected from 4 alternates to serve at a murder trial. Using the notation A_1A_3 , for example, to denote the simple event that alternates 1 and 3 are selected, list the 6 elements of the sample space S .

Solution

Choosing 2 from 4 gives

$$\binom{4}{2} = 6$$

outcomes:

$$S = \{A_1A_2, A_1A_3, A_1A_4, A_2A_3, A_2A_4, A_3A_4\}.$$

2.7

Four students are selected at random from a chemistry class and classified as male or female. List the elements of the sample space S_1 , using the letter M for male and F for female. Define a second sample space S_2 where the elements represent the number of females selected.

Solution

Each of the 4 selected students can be M or F , so

$$S_1 = \{MMMM, MMMF, MMFM, MFMM, FMMM, \\ MMFF, MFMF, MFFM, FMMF, FMFM, FFMM, \\ MFFF, FMFF, FFMF, FFFM, FFFF\}.$$

The second sample space records only the number of females:

$$S_2 = \{0, 1, 2, 3, 4\}.$$

2.8

For the sample space of Exercise 2.4,

- list the elements corresponding to the event A that the sum is greater than 8;
- list the elements corresponding to the event B that a 2 occurs on either die;
- list the elements corresponding to the event C that a number greater than 4 comes up on the green die;
- list the elements corresponding to the event $A \cap C$;
- list the elements corresponding to the event $A \cap B$;
- list the elements corresponding to the event $B \cap C$;
- construct a Venn diagram to illustrate the intersections and unions of the events A , B , and C .

Solution

The sample space is

$$S = \{(x, y) \mid x, y \in \{1, 2, 3, 4, 5, 6\}\},$$

where x is green and y is red.

- (a) Sum greater than 8:

$$A = \{(3, 6), (4, 5), (4, 6), (5, 4), (5, 5), (5, 6), (6, 3), (6, 4), (6, 5), (6, 6)\}.$$

- (b) A 2 on either die:

$$B = \{(2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), \\ (1, 2), (3, 2), (4, 2), (5, 2), (6, 2)\}.$$

- (c) Green die greater than 4 means $x = 5$ or $x = 6$:

$$C = \{(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}.$$

- (d)

$$A \cap C = \{(5, 4), (5, 5), (5, 6), (6, 3), (6, 4), (6, 5), (6, 6)\}.$$

- (e) Since sums greater than 8 cannot include a 2 on either die,

$$A \cap B = \emptyset.$$

- (f)

$$B \cap C = \{(5, 2), (6, 2)\}.$$

- (g) For the Venn diagram: place the 7 outcomes of $A \cap C$ in the overlap of A and C , the 2 outcomes of $B \cap C$ in the overlap of B and C , no outcomes in $A \cap B$, and no triple intersection.

2.9

For the sample space of Exercise 2.5,

- (a) list the elements corresponding to the event A that a number less than 3 occurs on the die;
- (b) list the elements corresponding to the event B that two tails occur;
- (c) list the elements corresponding to the event A' ;
- (d) list the elements corresponding to the event $A' \cap B$;
- (e) list the elements corresponding to the event $A \cup B$.

Solution

From Exercise 2.5,

$$S = \{1HH, 1HT, 1TH, 1TT, 2H, 2T, 3HH, 3HT, 3TH, 3TT, 4H, 4T, 5HH, 5HT, 5TH, 5TT, 6H, 6T\}.$$

- (a) Number less than 3 on the die means 1 or 2:

$$A = \{1HH, 1HT, 1TH, 1TT, 2H, 2T\}.$$

- (b) Two tails occur only for odd die values:

$$B = \{1TT, 3TT, 5TT\}.$$

- (c) Complement of A :

$$A' = \{3HH, 3HT, 3TH, 3TT, 4H, 4T, 5HH, 5HT, 5TH, 5TT, 6H, 6T\}.$$

- (d)

$$A' \cap B = \{3TT, 5TT\}.$$

- (e)

$$A \cup B = \{1HH, 1HT, 1TH, 1TT, 2H, 2T, 3TT, 5TT\}.$$

2.10

An engineering firm is hired to determine if certain waterways in Virginia are safe for fishing. Samples are taken from three rivers.

- (a) List the elements of a sample space S , using the letters F for safe to fish and N for not safe to fish.
- (b) List the elements of S corresponding to event E that at least two of the rivers are safe for fishing.
- (c) Define an event that has as its elements the points

$$\{FFF, NFF, FFN, NFN\}.$$

Solution

- (a) Each of the 3 rivers may be F or N , so

$$S = \{FFF, FFN, FNF, FNN, NFF, NFN, NNF, NNN\}.$$

- (b) At least two rivers are safe:

$$E = \{FFF, FFN, FNF, NFF\}.$$

- (c) The event

$$\{FFF, NFF, FFN, NFN\}$$

means that the second river is safe for fishing. So one possible definition is:

Event A = “the second river is safe for fishing.”

2.11

The resumés of two male applicants for a college teaching position in chemistry are placed in the same file as the resumés of two female applicants. Two positions become available, and the first, at the rank of assistant professor, is filled by selecting one of the four applicants at random. The second position, at the rank of instructor, is then filled by selecting at random one of the remaining three applicants. Using the notation M_2F_1 , for example, to denote the simple event that the first position is filled by the second male applicant and the second position is then filled by the first female applicant,

- list the elements of a sample space S ;
- list the elements of S corresponding to event A that the position of assistant professor is filled by a male applicant;
- list the elements of S corresponding to event B that exactly one of the two positions is filled by a male applicant;
- list the elements of S corresponding to event C that neither position is filled by a male applicant;
- list the elements of S corresponding to the event $A \cap B$;
- list the elements of S corresponding to the event $A \cup C$;
- construct a Venn diagram to illustrate the intersections and unions of the events A, B, C .

Solution

Let the applicants be M_1, M_2, F_1, F_2 . Since order matters,

$$S = \{M_1M_2, M_1F_1, M_1F_2, M_2M_1, M_2F_1, M_2F_2, \\ F_1M_1, F_1M_2, F_1F_2, F_2M_1, F_2M_2, F_2F_1\}.$$

- Listed above.
- First position filled by a male:

$$A = \{M_1M_2, M_1F_1, M_1F_2, M_2M_1, M_2F_1, M_2F_2\}.$$

- Exactly one position filled by a male:

$$B = \{M_1F_1, M_1F_2, M_2F_1, M_2F_2, F_1M_1, F_1M_2, F_2M_1, F_2M_2\}.$$

- Neither position filled by a male:

$$C = \{F_1F_2, F_2F_1\}.$$

-

$$A \cap B = \{M_1F_1, M_1F_2, M_2F_1, M_2F_2\}.$$

- Since A and C are disjoint,

$$A \cup C = \{M_1M_2, M_1F_1, M_1F_2, M_2M_1, M_2F_1, M_2F_2, F_1F_2, F_2F_1\}.$$

- In the Venn diagram, C is disjoint from A . Part of B overlaps A , and part of B lies outside A . Also $B \cap C = \emptyset$.

2.12

Exercise and diet are being studied as possible substitutes for medication to lower blood pressure. Three groups of subjects will be used to study the effect of exercise. Group 1 is sedentary, while group 2 walks and group 3 swims for 1 hour a day. Half of each of the three exercise groups will be on a salt-free diet. An additional group of subjects will not exercise or restrict their salt, but will take the standard medication. Use Z for sedentary, W for walker, S for swimmer, Y for salt, N for no salt, M for medication, and F for medication free.

- (a) Show all of the elements of the sample space S .
- (b) Given that A is the set of nonmedicated subjects and B is the set of walkers, list the elements of $A \cup B$.
- (c) List the elements of $A \cap B$.

Solution

The medication-free exercise groups are:

$$ZYF, ZNF, WYF, WNF, SYF, SNF.$$

The medication group is:

$$ZYM$$

(no exercise, salt, medication).

Thus

$$S = \{ZYF, ZNF, WYF, WNF, SYF, SNF, ZYM\}.$$

(a)

$$S = \{ZYF, ZNF, WYF, WNF, SYF, SNF, ZYM\}.$$

(b) Nonmedicated subjects:

$$A = \{ZYF, ZNF, WYF, WNF, SYF, SNF\}.$$

Walkers:

$$B = \{WYF, WNF\}.$$

Since $B \subseteq A$,

$$A \cup B = A = \{ZYF, ZNF, WYF, WNF, SYF, SNF\}.$$

(c)

$$A \cap B = \{WYF, WNF\}.$$

2.13

Construct a Venn diagram to illustrate the possible intersections and unions for the following events relative to the sample space consisting of all automobiles made in the United States.

$$F : \text{Four door}, \quad S : \text{Sun roof}, \quad P : \text{Power steering}.$$

Solution

Draw a three-circle Venn diagram inside the rectangle S (sample space of all automobiles made in the United States), with circles labeled F , S , and P . All pairwise intersections $F \cap S$, $F \cap P$, $S \cap P$, and the triple intersection $F \cap S \cap P$ are possible.

2.14

If

$$S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}, \quad A = \{0, 2, 4, 6, 8\}, \quad B = \{1, 3, 5, 7, 9\},$$

$$C = \{2, 3, 4, 5\}, \quad D = \{1, 6, 7\},$$

list the elements of the sets corresponding to the following events:

- (a) $A \cup C$;
- (b) $A \cap B$;
- (c) C' ;
- (d) $(C' \cap D) \cup B$;
- (e) $(S \cap C)'$;
- (f) $A \cap C \cap D'$.

Solution

(a)

$$A \cup C = \{0, 2, 3, 4, 5, 6, 8\}.$$

(b) A is even and B is odd, so

$$A \cap B = \emptyset.$$

(c)

$$C' = S \setminus C = \{0, 1, 6, 7, 8, 9\}.$$

(d) First,

$$C' \cap D = \{1, 6, 7\}.$$

Hence

$$(C' \cap D) \cup B = \{1, 3, 5, 6, 7, 9\}.$$

(e) Since $S \cap C = C$,

$$(S \cap C)' = C' = \{0, 1, 6, 7, 8, 9\}.$$

(f)

$$D' = S \setminus D = \{0, 2, 3, 4, 5, 8, 9\}.$$

Then

$$A \cap C = \{2, 4\},$$

so

$$A \cap C \cap D' = \{2, 4\}.$$

2.15

Consider the sample space

$$S = \{\text{copper, sodium, nitrogen, potassium, uranium, oxygen, zinc}\}$$

and the events

$$A = \{\text{copper, sodium, zinc}\}, \quad B = \{\text{sodium, nitrogen, potassium}\}, \quad C = \{\text{oxygen}\}.$$

List the elements of the sets corresponding to the following events:

- (a) A' ;
- (b) $A \cup C$;
- (c) $(A \cap B') \cup C'$;
- (d) $B' \cap C'$;
- (e) $A \cap B \cap C$;
- (f) $(A' \cup B') \cap (A' \cap C)$.

Solution

First compute complements in S :

$$A' = \{\text{nitrogen, potassium, uranium, oxygen}\},$$

$$B' = \{\text{copper, uranium, oxygen, zinc}\},$$

$$C' = \{\text{copper, sodium, nitrogen, potassium, uranium, zinc}\}.$$

(a)

$$A' = \{\text{nitrogen, potassium, uranium, oxygen}\}.$$

(b)

$$A \cup C = \{\text{copper, sodium, zinc, oxygen}\}.$$

(c) First

$$A \cap B' = \{\text{copper, zinc}\}.$$

Hence

$$(A \cap B') \cup C' = \{\text{copper, sodium, nitrogen, potassium, uranium, zinc}\}.$$

(d)

$$B' \cap C' = \{\text{copper, uranium, zinc}\}.$$

(e) Since $C = \{\text{oxygen}\}$ and oxygen is in neither A nor B ,

$$A \cap B \cap C = \emptyset.$$

(f) Since

$$A' \cap C = \{\text{oxygen}\},$$

and oxygen is also in $A' \cup B'$, we get

$$(A' \cup B') \cap (A' \cap C) = \{\text{oxygen}\}.$$

2.16

If

$$S = \{x \mid 0 < x < 12\}, \quad M = \{x \mid 1 < x < 9\}, \quad N = \{x \mid 0 < x < 5\},$$

find

- (a) $M \cup N$;
- (b) $M \cap N$;
- (c) $M' \cap N'$.

Solution

(a)

$$M \cup N = \{x \mid 0 < x < 9\}.$$

(b)

$$M \cap N = \{x \mid 1 < x < 5\}.$$

(c) Complements relative to S :

$$M' = \{x \mid 0 < x \leq 1 \text{ or } 9 \leq x < 12\},$$

$$N' = \{x \mid 5 \leq x < 12\}.$$

Therefore,

$$M' \cap N' = \{x \mid 9 \leq x < 12\}.$$

2.17

Let A, B , and C be events relative to the sample space S . Using Venn diagrams, shade the areas representing the following events:

- (a) $(A \cap B)'$;
- (b) $(A \cup B)'$;
- (c) $(A \cap C) \cup B$.

Solution

- (a) Shade all of S except the overlap $A \cap B$.
- (b) Shade the region outside both A and B .
- (c) Shade all of B together with the overlap $A \cap C$.

2.18

Which of the following pairs of events are mutually exclusive?

- (a) A golfer scoring the lowest 18-hole round in a 72-hole tournament and losing the tournament.
- (b) A poker player getting a flush (all cards in the same suit) and 3 of a kind on the same 5-card hand.
- (c) A mother giving birth to a baby girl and a set of twin daughters on the same day.
- (d) A chess player losing the last game and winning the match.

Solution

Two events are mutually exclusive if they cannot happen together.

- (a) Not mutually exclusive. A golfer may have the best single round and still lose the overall tournament.
- (b) Mutually exclusive. A flush requires all five cards same suit; 3 of a kind requires three cards same rank, which cannot occur in a flush.
- (c) Not mutually exclusive. A mother may give birth to twin daughters, which includes giving birth to a baby girl.
- (d) Not mutually exclusive. A player may lose the last game but still win the overall match.

Therefore,

Only (b) is mutually exclusive.

2.19

Suppose that a family is leaving on a summer vacation in their camper and that M is the event that they will experience mechanical problems, T is the event that they will receive a ticket for committing a traffic violation, and V is the event that they will arrive at a campsite with no vacancies. Referring to the Venn diagram of Figure 2.5, state in words the events represented by the following regions:

- (a) region 5;
- (b) region 3;
- (c) regions 1 and 2 together;
- (d) regions 4 and 7 together;
- (e) regions 3, 6, 7, and 8 together.

Solution

Using the standard numbering of the 3-set Venn diagram in Figure 2.5:

- region 1: $M \cap T \cap V$,
- region 2: $M \cap T \cap V'$,
- region 3: $M \cap T' \cap V$,
- region 4: $M \cap T' \cap V'$,
- region 5: $M' \cap T \cap V$,
- region 6: $M' \cap T \cap V'$,
- region 7: $M' \cap T' \cap V$,
- region 8: $M' \cap T' \cap V'$.

Hence:

- (a) Region 5:

$$M' \cap T \cap V.$$

In words: no mechanical problems, receives a traffic ticket, and arrives at a campsite with no vacancies.

- (b) Region 3:

$$M \cap T' \cap V.$$

In words: has mechanical problems, no traffic ticket, and arrives at a campsite with no vacancies.

- (c) Regions 1 and 2 together:

$$M \cap T.$$

In words: has mechanical problems and receives a traffic ticket.

- (d) Regions 4 and 7 together:

$$T' \cap V'.$$

In words: does not receive a traffic ticket and does not arrive at a campsite with no vacancies.

- (e) Regions 3, 6, 7, and 8 together:

$$T'.$$

In words: does not receive a traffic ticket.

2.20

Referring to Exercise 2.19 and the Venn diagram of Figure 2.5, list the numbers of the regions that represent the following events:

- (a) The family will experience no mechanical problems and will not receive a ticket for a traffic violation but will arrive at a campsite with no vacancies.
- (b) The family will experience both mechanical problems and trouble in locating a campsite with a vacancy but will not receive a ticket for a traffic violation.
- (c) The family will either have mechanical trouble or arrive at a campsite with no vacancies but will not receive a ticket for a traffic violation.
- (d) The family will not arrive at a campsite with no vacancies.

Solution

Using the same numbering as in Exercise 2.19:

- (a)

$$M' \cap T' \cap V$$

is region

$$\boxed{7}.$$

- (b) Mechanical problems and no vacancy, but no ticket:

$$M \cap T' \cap V$$

is region

$$\boxed{3}.$$

(c) Either mechanical trouble or no vacancies, but no ticket:

$$(M \cup V) \cap T'.$$

These are regions

$$\boxed{3, 4, 7}.$$

(d) Not arriving at a campsite with no vacancies means V' . These are regions

$$\boxed{2, 4, 6, 8}.$$

Exercises 2.21–2.31

2.21

Registrants at a large convention are offered 6 sightseeing tours on each of 3 days. In how many ways can a person arrange to go on a sightseeing tour planned by this convention?

Solution

Each day there are 6 tour choices, and the person selects one tour per day for 3 days.

By the multiplication principle:

$$6 \times 6 \times 6 = 6^3 = 216$$

$$\boxed{216 \text{ possible tour schedules}}$$

—

2.22

Patients are classified by:

$$AB^-, AB^+, A^-, A^+, B^-, B^+, O^-, O^+$$

and also according to blood pressure level:

low, normal, high

Solution

There are

8 blood types

and

3 pressure levels.

Thus,

$$8 \times 3 = 24$$

$$\boxed{24 \text{ possible classifications}}$$

—

2.23

An experiment consists of drawing a letter at random from the English alphabet.

Solution

The English alphabet contains

26

letters.

Therefore,

26 sample points

—

2.24

Students are classified by

Freshman, Sophomore, Junior, Senior

and

Male, Female.

Solution

There are

4 class levels

and

2 genders.

Hence

$$4 \times 2 = 8$$

8 possible classifications

—

2.25

Shoes come in 5 styles and each style has 3 display colors.

Solution

Number of possible shoes:

$$5 \times 3 = 15$$

15 different pairs

—

2.26

A person adopts 5 rules out of 7 health rules.

Solution

This is a combinations problem since order does not matter.

$$\begin{aligned}\binom{7}{5} &= \frac{7!}{5!2!} \\ &= \frac{7 \times 6}{2} = 21\end{aligned}$$

21 possible rule selections

2.27

A buyer can choose:

- 1 of 4 designs
- 1 of 3 heating systems
- Garage or carport
- Screened porch option

Solution

Choices:

$$\begin{aligned}4 \times 3 \times 2 \times 2 \\ = 48\end{aligned}$$

48 possible house plans

2.28

A drug may be purchased from

5 manufacturers

in

3 forms

and with

2 strengths.

Solution

Using the multiplication rule:

$$5 \times 3 \times 2 = 30$$

30 possible prescriptions

2.29

Each of 3 cars is tested with

5 gasoline brands

at

7 locations.

Solution

Total tests:

$$3 \times 5 \times 7$$

$$= 105$$

105 test runs

2.30

How many ways can a true-false test with 9 questions be answered?

Solution

Each question has

$$2$$

possible answers.

Therefore,

$$2^9 = 512$$

512 possible answer patterns

2.31

License numbers contain:

3 letters followed by 3 digits

with the first digit fixed as 5.

All digits must be different.

Solution

Letters

$$26^3$$

Digits

First digit fixed:

1 choice

Remaining digits must be different.

Second digit:

9 choices (0 – 9 except 5)

Third digit:

8 choices

Thus

$$26^3 \times 9 \times 8$$

$$= 17576 \times 72$$

$$= 1\,265\,472$$

1 265 472 possible license numbers

Exercises 2.54–2.64**2.54**

Let

$$P(T) = 0.6, \quad P(M) = 0.3, \quad P(T \cap M) = 0.15$$

where

T = invests in tax-free bonds, M = invests in mutual funds.

(a) Probability of investing in either

$$P(T \cup M) = P(T) + P(M) - P(T \cap M)$$

$$= 0.6 + 0.3 - 0.15$$

$$= 0.75$$

$$\boxed{P(T \cup M) = 0.75}$$

(b) Probability of neither

$$P((T \cup M)^c) = 1 - P(T \cup M)$$

$$= 1 - 0.75 = 0.25$$

$$\boxed{0.25}$$

—

2.55

Each item has

3 distinct letters

followed by

4 distinct nonzero digits.

Total possible codes:

$$P(26, 3) \times P(9, 4)$$

$$= (26 \cdot 25 \cdot 24)(9 \cdot 8 \cdot 7 \cdot 6)$$

Total codes:

$$15600 \times 3024 = 47\,174\,400$$

First letter vowel

5 choices

Remaining letters:

$$25 \times 24$$

Last digit even (nonzero):

$$\{2, 4, 6, 8\} \Rightarrow 4$$

Remaining digits:

$$8 \times 7 \times 6$$

Favorable:

$$(5 \cdot 25 \cdot 24)(4 \cdot 8 \cdot 7 \cdot 6)$$

$$= 3000 \times 1344$$

$$= 4\,032\,000$$

Probability

$$\frac{4\,032\,000}{47\,174\,400} = 0.0854$$

$$\boxed{0.0854}$$

—

2.56

$$P(B) = 0.25, \quad P(F) = 0.40, \quad P(B \cap F) = 0.15$$

(a) Probability defect in brakes or fuel

$$P(B \cup F) = P(B) + P(F) - P(B \cap F)$$

$$= 0.25 + 0.40 - 0.15$$

$$= 0.50$$

$$\boxed{0.50}$$

(b) Probability no defects

$$P((B \cup F)^c) = 1 - 0.50$$

$$= 0.50$$

$$\boxed{0.50}$$

—

2.57

Alphabet size:

$$26$$

(a) vowel except y

$$\{a, e, i, o, u\}$$

$$P = \frac{5}{26}$$

$$\boxed{\frac{5}{26}}$$

(b) before j

Letters before j:

 $a, b, c, d, e, f, g, h, i$

9

$$P = \frac{9}{26}$$

$$\boxed{\frac{9}{26}}$$

(c) after g

$$26 - 7 = 19$$

$$P = \frac{19}{26}$$

$$\boxed{\frac{19}{26}}$$

—

2.58

Total outcomes:

36

(a) sum = 8 $(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)$

5

$$P = \frac{5}{36}$$

$$\boxed{\frac{5}{36}}$$

(b) sum 5

Possible sums:

2, 3, 4, 5

Counts:

$$1 + 2 + 3 + 4 = 10$$

$$P = \frac{10}{36} = \frac{5}{18}$$

45

$$\frac{5}{18}$$

2.59

Total poker hands:

$$\binom{52}{5} = 2\,598\,960$$

(a) 3 aces

$$\binom{4}{3} \binom{48}{2}$$

$$= 4 \times 1128 = 4512$$

Probability:

$$\frac{4512}{2\,598\,960} = 0.001735$$

$$0.001735$$

(b) 4 hearts and 1 club

$$\binom{13}{4} \binom{13}{1}$$

$$= 715 \times 13 = 9295$$

$$P = \frac{9295}{2\,598\,960} = 0.00358$$

$$0.00358$$

2.61

Total students:

$$100$$

Math:

$$54$$

History:

$$69$$

Both:

$$35$$

$$46$$

(a) math or history

$$P(M \cup H) = \frac{54 + 69 - 35}{100}$$

$$= \frac{88}{100}$$

$$\boxed{0.88}$$

(b) neither

$$1 - 0.88 = 0.12$$

$$\boxed{0.12}$$

(c) history but not math

$$69 - 35 = 34$$

$$P = \frac{34}{100} = 0.34$$

$$\boxed{0.34}$$

—

2.62

Crust types:

$$3$$

Sauce types:

$$3$$

(a) combinations

$$3 \times 3 = 9$$

$$\boxed{9}$$

(b) plain thin crust + standard sauce

$$\frac{1}{9}$$

$$\boxed{\frac{1}{9}}$$

—

2.63

Given probabilities:

$$\text{Adult bedroom} = 0.03$$

$$\text{Child bedroom} = 0.15$$

$$\text{Other bedroom} = 0.14$$

$$\text{Office/den} = 0.40$$

$$\text{Other rooms} = 0.28$$

(a) PC in a bedroom

$$0.03 + 0.15 + 0.14$$

$$= 0.32$$

$$\boxed{0.32}$$

(b) not bedroom

$$1 - 0.32 = 0.68$$

$$\boxed{0.68}$$

(c) most likely location

Largest probability:

$$0.40$$

$$\boxed{\text{Office or den}}$$

—

2.64

Given:

$$P(X > 6000) = 0.42$$

$$P(X > 4000) = 0.04$$

(a)

$$P(X \leq 6000) = 1 - P(X > 6000)$$

$$= 1 - 0.42$$

$$= 0.58$$

$$\boxed{0.58}$$

(b)

$$P(X > 4000) = 0.04$$

$$\boxed{0.04}$$